DIC23
Decoder and Hierarchy Comparison

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Complete Lab 23 from PDF pages 91–93, including the 2-to-4 load comparison, 3-to-8 design and both 8-to-256 architectures, with every output checked.

2 Specification Mapping

S denotes simulated checks; D denotes analytical discussion.

Requirement and PDF pages	Deck / check	Status
NAND/NOR 2-to-4, 10fF (91–93)	D01/C0	S
Increasing load and stage-count question (93)	D01/C1–3	S+D
Design 3-to-8 (93)	D02; 8 codes	S
Two/three-stage 8-to-256 (93)	D03; 256 codes each	S
Hierarchical delay/power comparison (93)	D04; 10/100fF	S+D

3 Circuit and Design Derivation

The NAND 2-to-4 uses four NAND2 gates and four output inverters; the NOR version uses four NOR2 gates. Each has two input inverters, giving 28/20 MOS devices. The primary load is 10 fF per output. The 3-to-8 uses eight NAND3/inverter pairs and three input inverters, totaling 70 MOS devices.

The two-stage 8-to-256 separately decodes low/high 4-bit groups into 16 lines each, then combines them with 256 AND2 outputs. The three-stage design starts with four 2-to-4 groups, combines them into two 4-to-16 groups, then uses final AND2 gates. Stage count refers to logical decomposition; each positive AND uses NAND+inverter, giving four/six electrical gate levels excluding input inversion. Complemented-input paths add one inverter, giving five/seven levels. Counts including input inversions are 1872/1840 MOS. Intermediate four-bit combination gates are twice unit size to drive 16 final loads. Other unit $W_p/W_n/L = 3.6/1.8/0.25\,\mu$m; series NMOS/PMOS widths scale with stack length.

4 Simulation Setup

The model is supplied separately on the user's computer. These are 1 V, 27 °C teaching conditions; the handout's old-process 2.5 V numbers are not pass thresholds. Actual PrimeSim HSPICE S-2021.09 runs use `accurate`, `runlvl=5` and `.print/.measure` data. Every `.alter` resets baseline parameters and temperature. All cases completed without errors or failed measurements; model and convergence notices may remain. Public netlists retain only `* MODEL_SETUP_REQUIRED`, with no model path. Full data are under the lab's `results/`; large files are split by deck and indexed by `simulation_data.json`.

Four decks contain eight cases. The 2-to-4 repeats codes 0/1/2/3, 2 ns per code with 20 ps edges, at 10/50/100/500 fF and at most 1 ps steps. One-hot outputs are checked at 80% of each slot against 0.1/0.9 V rails. DATA DC sweeps print every input code and every output for the 3-to-8 and both 8-to-256 circuits. The hierarchical transient uses codes 0,1,0,15,255,0,16,0 at 5 ns/code with at most 2 ps steps; delay explicitly measures output1 for 0-to-1 and 1-to-0.

5 Results and Specification Checks

Load (fF)	NAND rise/fall (ps)	NOR rise/fall (ps)	NAND/NOR P (μW)	Fails
10	112.4/156.1	155.0/62.6	54.8/55.6	0
50	213.9/201.9	254.0/107.6	76.2/77.8	0
100	331.7/248.1	371.4/156.7	101.9/103.6	0
500	1259.8/396.7	1289.6/236.6	237.5/233.7	16

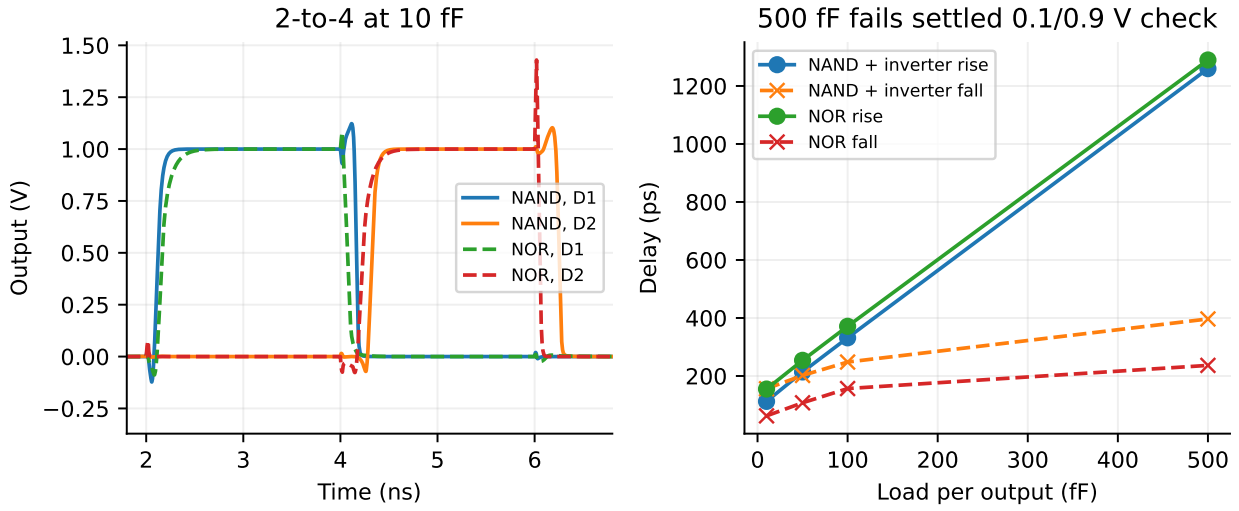


Figure 1: 2-to-4 waveforms and loading; 500 fF misses settled rail criteria at the specified 2 ns code width.

Load (fF)	Stages	t_{rise} (ps)	t_{fall} (ps)	P (μW)
10	2	744.7	590.3	522.7
10	3	1360.2	924.9	238.8
100	2	964.9	693.7	552.7
100	3	1579.8	1028.3	257.1

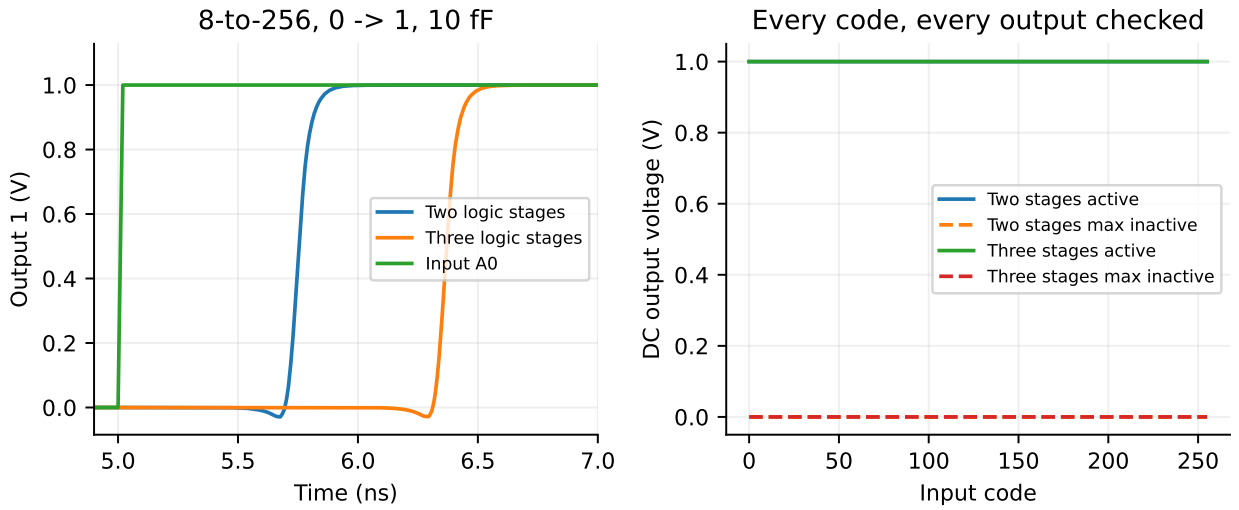


Figure 2: Representative 8-to-256 transition and exhaustive DC checks of every code and output.

All eight 3-to-8 codes pass. Each hierarchical design passes all 256 codes and every output, totaling 512 architecture/code checks. The large DC network prints convergence retries but ultimately completes and passes every truth check; no failed codes were removed.

The 2-to-4 passes all settled checks at 10/50/100 fF. At 500 fF, 16 of 18 architecture/slot checks miss the 0.1/0.9 V rails, so valid output at this code rate is not claimed. A measurable 50% propagation delay is different from reaching full logic rails.

6 Discussion

More stages do not invariably mean more delay at fixed load: added buffers can distribute fanout and improve transitions, but add intrinsic delay and capacitance. Here the 10 fF two-stage rise delay is about 744.7 ps versus 1360.2 ps for three stages. Three stages use less supply power for this stimulus, 238.8 versus 522.7 μW. These are not universal speed/power rules; sizing, stack resistance, capacitance, fanout and activity change the ordering.

Despite its extra level, the NAND 2-to-4 rises faster here, while NOR falls faster; one edge cannot summarize all timing. Simultaneous input-bit changes produce hazards; truth checks apply after settling, and timing plots retain spikes. No interconnect, extracted parasitics or layout is included, so device counts only describe topology cost.

7 Conclusion

All three decoder scales in the handout are complete, including exhaustive 256-code checks. Heavy-load failures and dynamic hazards remain explicit. Two/three-stage comparisons apply to the stated decomposition, sizes, loads and activity.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. Exact decoder primitives and the scalable hierarchy of `01_two_to_four_load.sp`, `02_three_to_eight_truth.sp`, `03_eight_to_256_truth.sp` and `04_eight_to_256_timing.sp`.

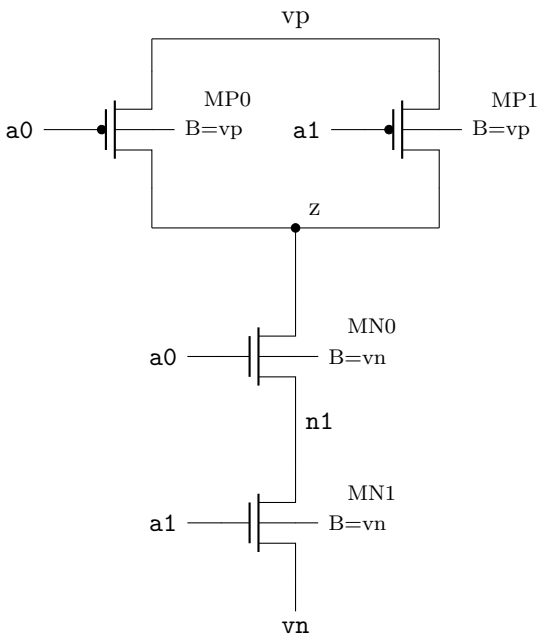


Figure 3: Exact NAND2 cell. AND2 connects NAND2.z to inv.a; its output is inv.z.

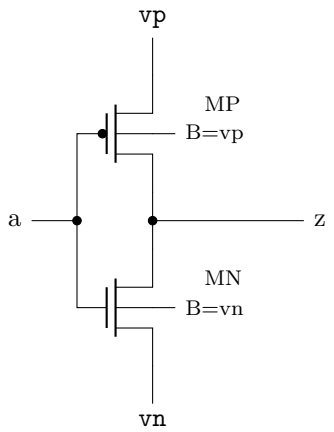


Figure 4: Exact inv cell used for input complements and NAND-output restoration.

All $L = 0.25\,\mu$m. inv uses $W_P/W_N = 3.6k/1.8k\,\mu$m; nand2 uses $3.6k/3.6k\,\mu$m. PMOS sources/bodies connect to vp, NMOS bodies to vn. Named n1 is the internal NMOS stack node, not a supply rail.

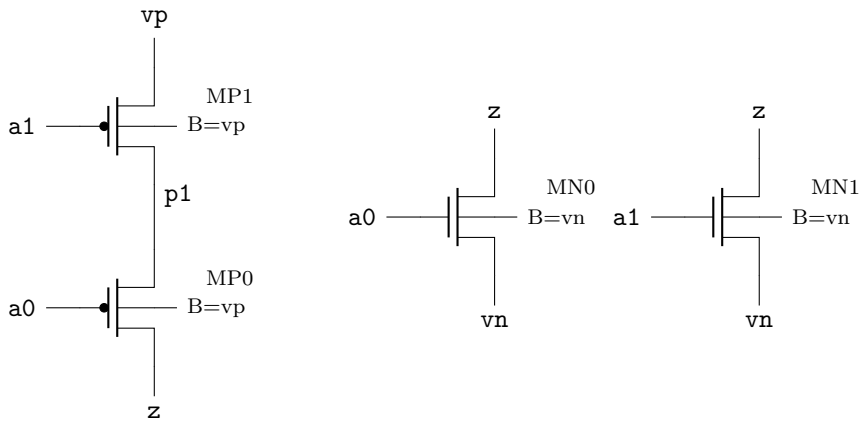


Figure 5: Exact NOR2 alternative: MP1 lies above MP0 in the PMOS stack.

NOR2 uses $W_P = 7.2k\,\mu$m and $W_N = 1.8k\,\mu$m. MP0's source is p1, while its body remains vp. For the 2-to-4 NAND version, each positive minterm is NAND2 followed by inv. With input order $(a1, a0)$, the NAND input pairs for outputs 0–3 are $(nb1, nb0)$, $(nb1, a0)$, $(a1, nb0)$ and $(a1, a0)$. The NOR alternative uses $(a1, a0)$, $(a1, rb0)$, $(rb1, a0)$ and $(rb1, rb0)$ respectively. nb/rb are generated by the architecture's own two input inverters; vp maps to supply vn or vr and the cell's vn maps to global 0. Both variants attach cl from every output to 0.



Figure 6: 3-to-8 decoder: all eight minterms are explicit NAND3 + inv cells.

nand3 extends the shown NAND2 pattern with three parallel PMOS and three series NMOS; $W_P/W_N = 3.6k/5.4k\,\mu$m. All k=1; 3 input inverters plus 8 AND3 cells total 70 MOS, each output loaded by 10 fF. The full 8-to-256 topology is shown on the next page without drawing 256 duplicate output cells.

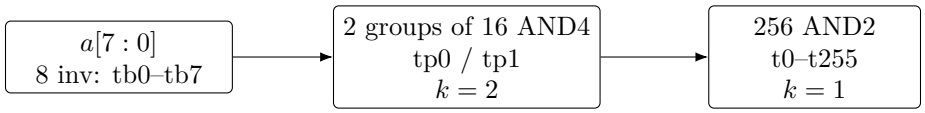


Figure 7: Two logical stages after input complement generation. Each AND is NAND + inv.

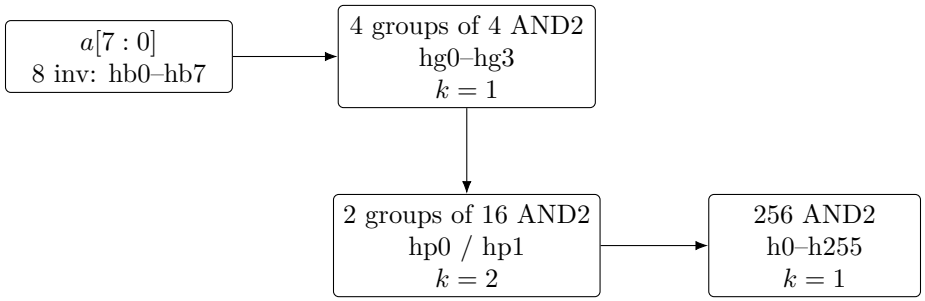


Figure 8: Three logical stages, with exact grouping and sizing multipliers.

For integer output index j, the two-stage path combines tp0[$j \bmod 16$] (bits 3:0) and tp1[$\lfloor j/16 \rfloor$] (bits 7:4). The three-stage path first decodes bit pairs (1:0), (3:2), (5:4), (7:6), combines adjacent pairs into hp0/hp1, then uses the same low/high nibble indices. Every AND2 or AND4 is the exact NAND-plus-inverter combination. nand4 uses four parallel PMOS ($3.6k\,\mu$m) and four series NMOS ($7.2k\,\mu$m), all $L = 0.25\,\mu$m; k also scales the following inverter. PMOS body rails map to vt for the two-stage design and vh for the three-stage design; all NMOS bodies connect to 0. Every output has cl=10 fF at baseline. Counts including eight input inverters are 1872 and 1840 MOS. The 2/3 logical stages contain 4/6 electrical gate levels after the input literals; complemented-input paths add one level, giving 5/7. Supply/input sources and repeated capacitors are omitted from the hierarchy boxes but retained in the exact source decks.